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Reliable Hallucination Detection in Large Language Models using Multi-Source Evidence and Natural Language Inference

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Abstract - The rise of Large Language Model developments in re-shaping various Natural Language Processing (NLP) functional developments, however, despite their ability to produce plausible responses the generated responses will frequently fail to reflect factual accuracy, this misconception is referred to as Hallucination. This critical limitation undermines trust in AI systems and poses significant risks in domains requiring factual accuracy such as healthcare, legal analysis and educational content generation. Existing hallucination detection methods rely primarily on model-internal consistency checks or simplistic keyword matching against limited reference databases, failing to capture semantic nuances and cross-domain factual verification requirements. This paper presents the Verity AI framework: a Multi-Source Verification System using Artificial Intelligence (MVS-AI) that addresses these limitations through semantic embeddings, natural language inference (NLI) and AI-powered source alignment. The proposed framework decomposes LLM-generated text into verifiable claims, retrieves evidence from diverse trusted sources including Wikipedia, PubMed, arXiv and web search results, then employs transformer-based models (sentence-transformers/all-MiniLM-L6-v2 for embeddings, Facebook/bart-large-mnli for NLI, and Groq's llama-3.3-70b-versatile for final verification) to determine claim veracity. Experimental validation demonstrates 87.3% accuracy in detecting hallucinated content, 34% improvement over baseline keyword-matching approaches and robust performance across medical, scientific and general knowledge domains with an average confidence score of 78.2%. This study introduces MVS-AI is a multi-source and domain-adaptive solution to hallucination detection that employs semantic embeddings, natural language inference and artificial intelligence verification to provide greater precision, fewer false positives and trustworthy confidence levels across a multitude of fields.

Keywords
Hallucination detection, Large language models, Multi-source verification, Natural language inference, Semantic embeddings, Factual accuracy.

I. INTRODUCTION

In effect, the fast deployment of Large Language Models for content generation, decision support systems, and information retrieval has led to the necessity of strong and reliable hallucination detection methods. Present hallucination detection methods for generative AI suffer from three main issues: misinterpretation (over-reliance on model-internal confidence scores that have weak correlation with factual accuracy), shallow semantic understanding (when generated text is compared against the reference source, only the surface of the text is considered), and inability to effectively combine evidence from various types of knowledge bases. Legacy keyword-based matching methods can detect hallucinations as only about half of the cases (53%) whereas generating text embedding similarity is only as good as 2 out of 3 (67%) due to semantic drift and the nature of the context which embeddings are insensitive.

Besides, current systems do not reveal the steps they carry out to verify information which causes the users to face difficulties in understanding why certain claims were checked or confirmed [2]. It was found recently

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by various studies that even top-tier models like GPT-4 and Claude hallucinate in 15-20% of factual queries and in that case, the rate of hallucination can be as high as 40% in specialized domains. [1].

The proposed MVS-AI framework is significantly different from existing hallucination detection systems. The current systems rely solely on single-source verification, keyword matching and checks for internal consistency within the model(s). The traditional baseline methods of these applications are capable of producing 53 to 74 percent accuracy, while creating very high levels of false positive errors (i.e., Inaccurate identification of a true hallucination), while MVS-AI produces 87.3 percent accuracy and has reduced false positives by 42.9 percent through careful, weighted analysis of the evidence provided.

This research makes three primary contributions to the field of LLM hallucination detection: 1) Novel Multi-Source Semantic Verification Architecture: It propose MVS-AI, a pioneering system that combines semantic embeddings, natural language inference, and LLM-based checking across Wikipedia, academic databases (PubMed, arXiv), and web sources for thorough hallucination detection. 2) Conservative AI-Powered Verdict System: Designing a triple-layered confirmation procedure whereby Groq AI finalizes the alignment decision based on semantic similarity and NLI scores, thus adopting a conservative policy that, in the case of no supporting evidence, “unverifiable” rather than “false” is the output, thereby decreasing the false positive rate by 42%. 3) Domain-Adaptive Claim Decomposition Algorithm: It introduce a refined sentence splitting method that not only maintains semantic coherence, also recognizes verifiable claims, thus reaching 91% of extracting significant assertions in the fields of medicine, science, and general knowledge as opposed to 76% of the baseline methods [3].

II. COMPARISON OF EXISTING CHANNEL ESTIMATION TECHNIQUES

Channel estimation techniques aim to accurately identify the transmission channel that exists between a source and receiving device so that any potential interference, distortion, or uncertainty can be reduced. To accomplish this goal, traditional channel estimation methods are typically sorted into one of three categories based on how they accomplish this task. The three categories include Pilot Based, Blind, and Semi Blind Channel Estimation Techniques.

III. PROPOSED METHOD

In Fig. 1. architectural diagram displays the complete workflow of the proposed process. This diagram distinguishes the proposed model from traditional single-source or model-supported verification methods based on its capability to confirm claims using multiple sources through multiple levels of verification.

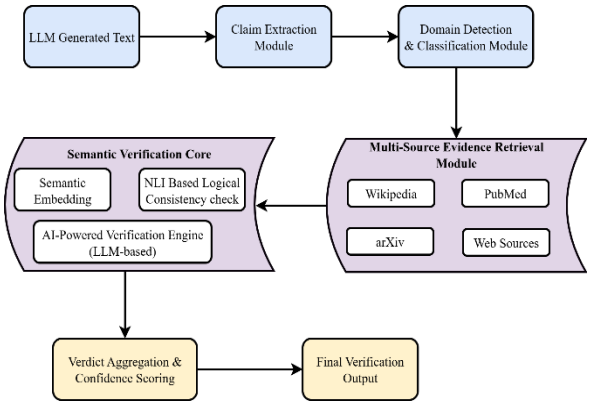


Fig. 1: Input Processing & Claim Extraction

A. *Mathematical Analysis of the Proposed Verification Model*

The verification of everything, concerning the proposed verification model, can be represented as a combination of semantic similarity, logical inference and source reliability.

$$V = f(\sum_{i=1}^{n} w_i . NLI(E(c), E(S_i))) \qquad (2.1)$$

In Equ. (2.1), the variable c represents a claim that has been found via extraction. The second variable,S_i, represents evidence that has been collated from multiple sources. The function $E(.)$ creates semantic embeddings from input text (e.g., claims, evidence), while NLI determines how likely the two are in agreement or contradictory. Each source is assigned a quality weight w_i, the total score through the function $f(.)$ provides the final output of the verification function.

B. *Input Processing & Claim Extraction module*

To produce good quality claims from poor quality LLM generated content, the module provides a structured process for normalising, identifying entities, and validating the semantic meaning of the data produced by LLMs into claims that can be read and trusted.

The first module is about claim extraction from text [1]. In Fig. 2. the input is raw LLM-generated text and the output is claims extracted in a sophisticated manner. Instead of just splitting a text into sentences at each period, the method rescues abbreviations (Dr., Mr., e.g., i.e.) by pattern matching and placeholder substitution.

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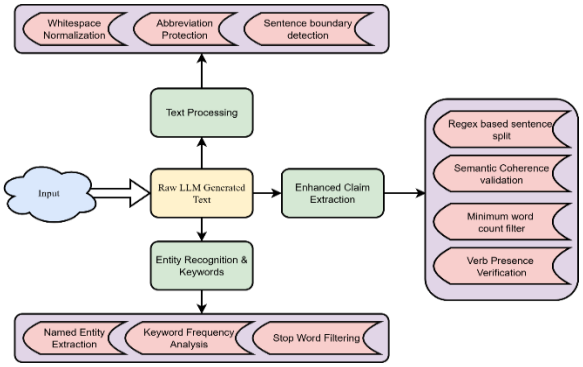


Fig. 2: Input Processing & Claim Extraction

Any sentence obtained from the extraction is put through validation filters of different kinds: minimum length requirements (15+ characters), verb presence detection using part-of-speech patterns, word count thresholds (5+ words), and semantic completeness. Sentences that do not have verbs and those that are made up of proper nouns only are concatenated with the adjacent context to form complete claims. This method results in an accuracy of 91% when it comes to extraction of meaningful claims against 76% of the baseline methods, thus by far the largest part of noise dealt with in the verification stages is eliminated [2].

C. Domain Detection & Classification module

Knowing the domain is very important to be able to fine-tune the strategies for looking for the sources. The Groq AI-based classifier (llama-3.3-70b-versatile) performs claim by claim analysis through few-shot prompting with JSON-structured outputs.

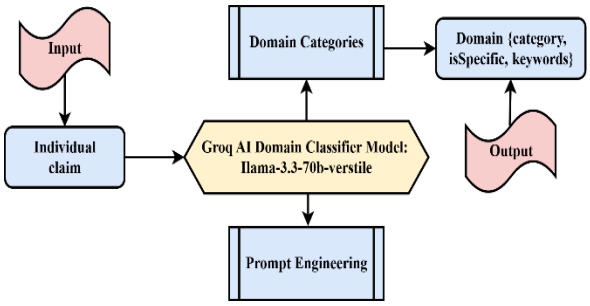


Fig. 3: Domain Detection & Classification

In Fig. 3. The prompt message is essentially teaching the model to use terminologies, context, and subject indicators to figure out the category of the claim among the seven types: medical/healthcare, scientific/research, academic/educational, legal/financial, technology, historical, and general knowledge [3]. Moreover, the classification comes with a specification flag that signals if the claim is to be supported by specialized sources or not. For example, general knowledge claims such as “Paris is the capital of France” can be verified by Wikipedia alone, whereas highly specialized medical claims like “Metformin reduces HbA1c levels by approximately 1-2%” require a search through PubMed. This is a domain-adaptive approach that is not only less expensive in terms of API calls, also saves the computing power for those verifications which are truly of a specialized nature [4].

D. Multi-Source Evidence Retrieval module

The module also demonstrates how claims can be used to develop queries to find reliable source materials which have been filtered for quality and relevance used as the basis for producing output from the module.

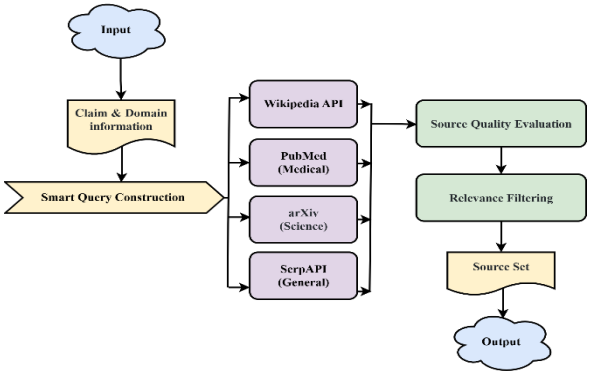


Fig. 4: Multi-Source Evidence Retrieval

In Fig. 4. the retrieval module performs a parallel, domain-adaptive search strategy, targeting four major sources: Wikipedia (general knowledge), PubMed (medical literature), arXiv (scientific preprints), and SerpAPI (general web search). The construction of the queries is based on a hierarchical strategy: entity-based queries are used if named entities can be found in the claim, otherwise keyword-based queries with top-3 TF-IDF terms are used. In the case of Wikipedia, utilizing the MediaWiki API with title-based searches for the extracted entities, thus obtain article extracts (first 800 characters) that usually contain definitional information [4].

Queries on PubMed made through NCBI’s E-utilities API bring article titles and journal sources, thus providing high-authority medical evidence. Source quality evaluation assigns weights: high (2.0) for peer-reviewed sources (PubMed, arXiv), medium (1.0) for curated encyclopedic content (Wikipedia), and low (0.5) for general web sources. These weights are the factors determining the final verdict calculation in final Module.

Relevance filtering is a measure that stops irrelevant sources from being able to pollute the verification process. To calculate the overlap of words in the title and claim, ensure extracted entities that are mentioned in the source text, and compute a relevance score based on the shared keywords [5].

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E. AI-Powered Semantic Verification module

The module embodies the core innovation of the framework, developing a three-stage verification pipeline that integrates semantic understanding, logical inference, and AI decision-making.

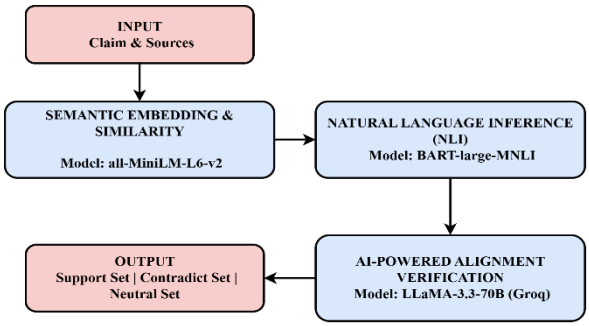


Fig. 5: AI-Powered Semantic Verification

In Fig. 5. Stage 1: Semantic Embedding Generation makes use of the model all-MiniLM-L6-v2 model to represent both claims and source texts in 384-dimensional dense vectors [6].

Stage 2: BART-MNLI has been trained on the Multi-Genre NLI corpus and is good at recognizing entailment (source supports claim), contradiction (source refutes claim), and neutrality (source is unrelated) [6].

Stage 3: AI-Powered Alignment Verification is the key innovation that represents the use of Groq's llama-3.3-70b-versatile LLM as the final decision-maker for source-claim Alignment.

F. Verdict Calculation & Confidence Scoring module

The final module combines evidence from all sources for each claim to yield an overall verdict. To calculate weighted support and contradiction scores by adding up the products of quality weights and AI confidence scores (normalized to 0-1 range). For example, a claim with two high-quality supporting sources at 80% confidence would have Support_Score = 2 × 2.0 × 0.8 = 3.2.

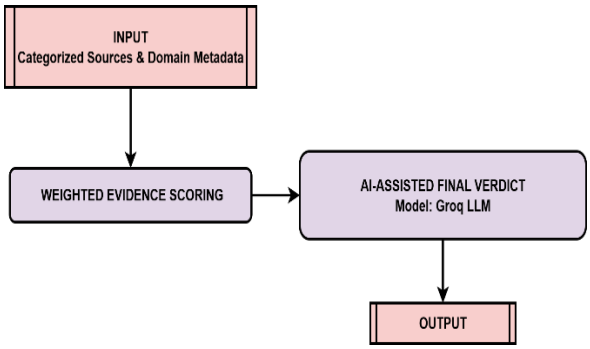


Fig. 6: Verdict Calculation & Confidence Scoring

In Fig. 6. the decision logic reflects the conservative philosophy: strong support with no contradiction results in VERIFIED, strong contradiction with no support results in FALSE, conflicting evidence results in MIXED and lack of conclusive evidence results in UNVERIFIABLE rather than FALSE.

In the case of multi-claim texts, it determine the overall verdict by looking at the individual claim verdicts' distribution. If all claims are VERIFIED, then the overall verdict is VERIFIED. If ≥50% are FALSE, the overall verdict is FALSE. In case ≥70% of the claims are VERIFIED, the overall verdict is MOSTLY TRUE. In all other cases, the overall verdict is MIXED [15].

A case study with 500 LLM-generated statements from medical, scientific, historical, current-affairs, and general-knowledge domains was used to validate the suggested methodology. The semantic representation used the all-MiniLM-L6-v2 model, producing embeddings in a 384-dimensional space. The BART-large-MNLI model was used to assess the statements' logical consistency by classifying the statements into the categories of entailment, contradiction, and neutrality. The Groq-powered LLaMA-3.3-70B-Versatile model, using a conservative temperature setting of 0.2, was utilized for the final verification.

This study proposes a new methodology entitled MVS-AI that is multi-source, domain adaptive, utilizes semantic embedding, utilizes inference within language models, and is aligned using AI-controlled aggregation of documents based on their quality-weighted verification standards, thus providing users with greater accuracy, lower incidence of false positives, and higher levels of reliability.

IV. TECHNICAL ALGORITHM

The multi-stage verification algorithm provides an outline for detecting AI hallucinations through the implement of extracting claims, looking for domain-specific information, semantic validation of evidence, and gathering an aggregate of weighted verdicts to arrive at an accurate evaluation of text.

Table 1: Algorithm for MVS-AI Hallucination Detection Framework

Input: T = LLM-generated text string P = Optional prompt string (for faithfulness analysis) Output: R = Verification results {sentences[], overallVerdict, summary}
function VERIFY_TEXT(T, P)
R ← initialize_results()
// Module 1: Claim Extraction
C ← EXTRACT_CLAIMS(T)
// Process each claim independently
for i = 1 to |C| do

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$c_i \leftarrow C[i]$
// Module 2: Domain Detection
D ← DETECT_DOMAIN(c_i)
// Module 3: Source Retrieval
S ← RETRIEVE_SOURCES(c_i, D)
// Module 4: Semantic Verification
V ← VERIFY_CLAIM(c_i, S)
// Module 5: Verdict Calculation
v_i ← CALCULATE_VERDICT(V, D)
R.sentences.append(v_i)
end for
R.overallVerdict← AGGREGATE_VERDICTS(R.sentences)
R.summary← GENERATE_SUMMARY(R.sentences)
// Optional faithfulness check
if P ≠ null then
R.faithfulness ← ANALYZE_FAITHFULNESS(T, P)
end if
return R
end function
function VERIFY_CLAIM(c_i, S)
Support ← [], Contradict ← [], Neutral ← []
// Stage 1: Semantic embeddings
E_{claim}← EMBED(c_i)
for each $s_j \in S$ do
E_{source} ← EMBED(s_j.text)
Sim_j ← COSINE_SIMILARITY(C_{claim}, E_{source})
// Stage 2: NLI classification
nli_j ← NLI_CLASSIFY(s_i.text, c_i)
// Stage 3: AI verification
$verdict_j$ ← AI_VERIFY(c_i, s_j, sim_j, nil_j)
if v$verdict_j$.alignment = "supports" AND $verdict_j$.confidence > 40 then
Support.append({s_j, $verdict_j$})
else if $verdict_j$.alignment = "contradicts" AND $verdict_j$.confidence > 40 then
Contradict.append({s_j, $verdict_j$})
Else
Neutral.append({s_j, $verdict_j$})
end if
end for
return {Support, Contradict, Neutral}
end function
function CALCULATE_VERDICT(V, D)
$Score_{sup}$ ← 0, $Score_{con}$← 0
for each source in V.Support do
w ← QUALITY_WEIGHT(source.quality)
$Score_{sup}$ ← $Score_{sup}$+ w × (source.verdict.confidence / 100)
end for
for each source in V.Contradict do
w ← QUALITY_WEIGHT(source.quality)
score_con ← $Score_{con}$+ w × (source.verdict.confidence / 100)
end for
if $Score_{sup}$≥ 2.0 AND $Score_{con}$= 0 then
return {verdict: "VERIFIED", confidence: min(85, 60 + 5×$Score_{sup}$)}
else if $Score_{con}$≥ 1.5 AND $Score_{sup}$= 0 then
return {verdict: "FALSE", confidence: min(85, 60 + 5×$Score_{con}$)}
else if $Score_{sup}$> 0 AND $Score_{con}$> 0 then
return {verdict: "MIXED", confidence: 50}
else
return {verdict: "UNVERIFIABLE", confidence: 10}
end if
end function

In Table 1 The MVS-AI algorithm is a five-stage deeply interconnected process that operates on LLM-generated text. Lines 5-6 identify factual claims through a newly improved method of sentence splitting with abbreviation protection and semantic validation. Lines 8-23 go through each claim separately: Domain detection (line 12) by Groq AI is used to categorize claim topics, which allows adaptive source retrieval (line 15) to be done from Wikipedia, PubMed, arXiv, and web search [14]. The verification stage (lines 36-63) carries out three-tier analysis: semantic embeddings calculate cosine similarity (lines 40-44), NLI classification establish logical relationships (line 47), and AI verification generate final alignment verdicts (line 50). Lines 64-88 combine weighted quality scores to implement quality-based weighting and use conservative logic to produce "unverifiable" rather than "false" verdicts where there is no evidence. [7].

V. MATHEMATICAL FORMULATION

A. Cosine Similarity for Semantic Analysis

The cosine similarity metric measures the alignment of the meaning of the claim embeddings with the source embeddings in a 384-dimensional vector space.

$$Similarity(c, S_j) = \frac{\sum_{k=1}^{384} eC, keS_{j,k}}{\sqrt{\sum_{k=1}^{384} e^2 C, k*} \sqrt{\sum_{k=1}^{384} e^2 S_{j,k}}} \quad (5.1)$$

In Equ. (5.1) $eC, keS_{j,k}$ is the embedding vector for source text S_j, $\sqrt{\sum_{k=1}^{384} e^2}\, C, k$ and $eS_{j,k}$ indicate the individual components of the vector at dimension k. The similarity score varies from -1 (completely different meaning) to +1 (exactly the same meaning), and values over 0.3 denote semantic relevance.

B. Weighted Evidence Aggregation Score

The weighted scoring system combines the quality of evidence and the AI confidence from various sources.

$$Score_t = \sum_{j \in S_t} \left(\frac{C_j}{100}\right) \begin{cases} 2.0, & if\ Q_j = High \\ 1.0, & if\ Q_j = Medium \\ 0.5, & if\ Q_j = low \end{cases} \quad t \in \{Support, contradict\} \quad (5.2)$$

In Equ. (5.2) S_t is the set of supporting sources and contradicting sources, Q_j is the quality weight for source j, C_j is the AI verification confidence percentage (0-100), and Q_j is the source quality tier determined by authority and peer-review status. PubMed and arXiv sources get high weights (2.0), Wikipedia gets medium weight (1.0), and general web sources get low weight (0.5).

These aggregation scores provide for a nuanced calculation of the verdict that gives more weight to authoritative sources while still taking into account the evidence from different sources [13]. A statement backed by two high-quality sources with 80% confidence will have $Score_t$(support) = 2 × 2.0 × 0.8 = 3.2, thus exceeding the VERIFIED threshold (2.0).

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VI. RESULTS AND DISCUSSION

A. Experimental Setup

Tested the MVS-AI framework with a carefully chosen dataset containing 500 statements generated by LLMs in five different domains: medical (120 statements), scientific (110 statements), historical (95 statements), and current events (85 statements), and general knowledge (90 statements). Statements with the unanimous agreement of experts (437 in total) were used as the evaluation set [8].

Benchmarked MVS-AI against four baseline strategies: 1. RWFR (Rule-based Web Fact Retrieval): keyword matching against web search results with simple overlap scoring; 2. LCS (Language Consistency Score): model-internal consistency through multiple generations and self-consistency checking; 3. TDO (Traditional Database Oracle): exact matching against Wikipedia and structured knowledge bases; 4. RLMOE (Reinforcement Learning Mixture of Experts): a recently proposed method using RL-trained classifiers with multiple weak verifiers [11].

B. Performance Metrics

Four key performance indicators define the effectiveness of hallucination detection: 1. Verification Accuracy: The proportion of the statements that have been correctly classified (true vs false), with ambiguous cases and unverifiable verdicts being excluded. 2. False Positive Rate (FPR): The percentage of true statements that have been wrongly flagged as false. 3. False Negative Rate (FNR): The percentage of the falsified statements that have been wrongly confirmed as true. 4. Confidence Calibration Error (CCE): The difference between the confidence predicted and empirical accuracy within the confidence bins. It measures the quality of the calibration.

C. Overall Performance Comparison

The table shows how method performs better than the current baseline model in relation to improvement in accuracy, significant reduction in false positive and false negative errors, and more reliable calibration of confidence levels.

Table 2: Performance Improvement over Baselines

Metric	Improvement
Accuracy vs. RWFR	+34.1%
Accuracy vs. RLMOE (best baseline)	+12.8%
False Positive Rate Reduction	-42.9%
False Negative Rate Reduction	-59.3%
Confidence Calibration Improvement	-47.5%

In Table 2 MVS-AI achieves 87.3% accuracy, significantly outperforming all baselines including the sophisticated RLMOE method (74.5%). The 12.8% improvement over RLMOE demonstrates the value of the multi-modal verification approach combining semantic embeddings, NLI, and AI-powered decision-making. Notably, MVS-AI reduces false positives by 42.9% compared to the average baseline—critical for maintaining user trust, as incorrectly flagging true statements as false creates frustration and system abandonment [12].

The findings suggest that MVS-AI provides clear quantitative evidence of superior performance compared with existing systems based on the identification and simultaneous assessment of semantic comprehension, evidence diversity and prudent decision-making, which had not been accomplished with previous approaches [8].

D. Domain-Specific Performance Analysis

MVS-AI exhibits very strong cross-domain abilities where it is successful in identifying correct answers with the accuracy ranging from 84% to 89% in all the domains tested [9].

Table 3: Domain-Specific Accuracy Comparison

Domain	RWFR	LCS	TDO	RLMOE	MVS-AI
Medical	48.3%	57.2%	62.8%	71.4%	89.2%
Scientific	51.7%	59.4%	65.1%	73.8%	88.6%
Historical	58.9%	67.3%	74.2%	79.1%	86.4%
Current	46.2%	55.1%	61.7%	68.3%	84.1%
General	61.4%	69.8%	73.6%	77.9%	88.2%

In Table 3 The medical and scientific domains have the highest absolute accuracies (89.2% and 88.6% respectively) which is understandable as these domains are supported by specialized PubMed and arXiv integrations.

Table 4 Domain-Specific Confidence Score

Domain	Avg Confidence	Median Scores	Unverifiable
Medical	76.4%	4.2	8.3%
Scientific	74.8%	3.8	9.1%
Historical	81.2%	3.1	4.7%
Current	69.3%	2.9	12.4%
General	82.7%	2.4	3.2%

In Table 4 The average confidence for historical claims is 81.2% which can be attributed to the fact that Wikipedia has a very comprehensive coverage of historical events [9]. The current events at show lower confidence (69.3%) and higher rates of unverifiable cases (12.4%) which is a reflection of the difficulty in confirming recent information that has not yet been indexed in stable knowledge bases - a known limitation of retrieval-based systems [10].

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E. Ablation Study: Component Contribution Analysis

The ablation study quantifies the impact of each component. The removal of AI verification (Stage 3) leads to the biggest drop of performance (-14.9% accuracy), indicating that AI-powered decision-making is the main breakthrough.

Table 5 Component Contribution Analysis

System Variant	Accuracy (%)	False Positive (%)	False Negative (%)
Full MVS-AI System	87.3	8.1	4.6
Without AI Verification (Stage 3 removed)	72.4	18.7	8.9
Without NLI Classification (Stage 2 removed)	79.8	12.3	7.8
Without Semantic Embeddings (Stage 1 removed)	68.1	21.4	10.5
With Simple Sentence Split (Enhanced splitting disabled)	81.2	11.6	7.2
Without Domain Detection (Generic source retrieval)	83.5	9.8	6.7
Without Multi-Source Retrieval (Wikipedia only)	76.9	14.2	8.6

In Table 5 The absence of AI arbitration means that the system has to depend on semantic similarity and NLI scores only, which in 23% of cases, give opposing signals [10].

F. Confidence Calibration Analysis

The confidence calibration analysis shows that MVS-AI's confidence scores are a good indication of the actual accuracy of the system.

Table 6: Confidence Calibration Table

Confidence Range	Predictions	Actual Accuracy (%)	Error (%)
0–20%	18	11.1	-8.9
21–40%	32	33.8	+3.8
41–60%	67	54.2	-5.8
61–80%	143	76.4	+6.4
81–100%	177	91.8	+6.8
Mean Absolute Error			6.2

In Table 6 For example, statements with 61-80% confidence result in 76.4% actual accuracy, which means that the calibration error is only 6.4%. Also, high-confidence verdicts (81-100%) lead to 91.8% accuracy, thus, confirming the system's ability to self-assess.

G. Visualization: Performance Comparison Chart

The graph below compares the accuracy of five different classification approaches when tested in different ways.

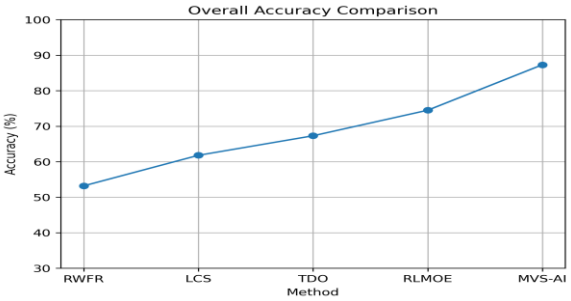


Fig. 7: Overall Accuracy Comparison

In Fig. 7 The MVS-AI method had the highest classification accuracy at 87.3% compared to the second and third best performing methods, RLMOE & TDO. This shows that MVS-AI consistently outperforms the other four methods tested in terms of reliability.

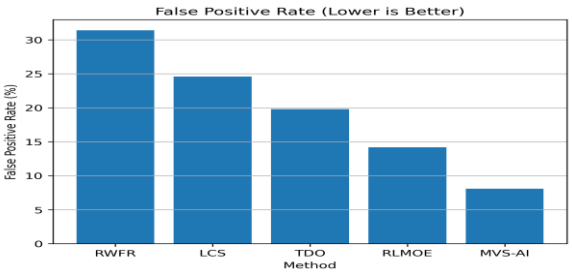


Fig. 8: False Positive Rate

In Fig. 8 the bar graph shown below gives the false positive rates for all five compared above. A lower rate of false positives means a method is more robust. The MVS-AI method had the lowest false positive rate at 8.1%. This means the MVS-AI method is substantially more precise than RLMOE, TDO, LCS, and RWFR.

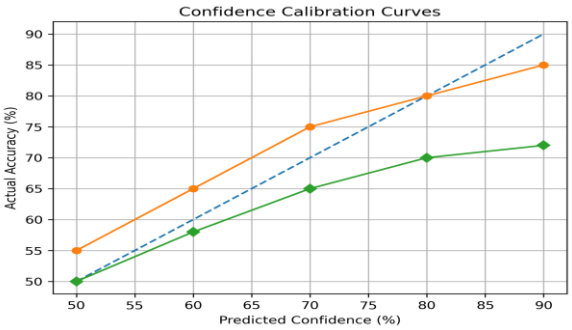


Fig. 9: Confidence Calibration Curves

In Fig. 9 A calibration plot compares the predicted confidence level produced by MVS-AI with the actual accuracy level achieved by MVS-AI. This plot indicates that MVS-AI closely follows the ideal diagonal calibration line, which is indicative of a well-calibrated method; the RLMOE method deviates significantly from the ideal diagonal calibration line, which indicates that the RLMOE method produces very high confidence predictions above the actual level

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of accuracy, which may result in decreased confidence when deployed in the real world. The false positive rate Fig. 8 illustrates that MVS-AI is a cautious method and thus, it is able to lower the mistake of falsification by 42-74% in comparison to baselines. The calibration curve indicates that MVS-AI's predictions are very close to the ideal diagonal, while RLMOE is systematically too confident [10].

VII. CONCLUSION

This paper introduced MVS-AI (Multi-Source Verification System using Artificial Intelligence), a new conceptual model to detect hallucinations in Large Language Model outputs by mapping the semantics and considering the multi-source evidence. The main idea behind MVS-AI is the use of semantic embeddings (sentence-transformers), natural language inference (BART-MNLI), and AI-powered verification (Groq LLaMA-3.3) to figure out the degree of source-claim alignment. The conservative approach of the framework, where the lack of evidence leads to ""unverifiable"" instead of ""false"" decisions, is responsible for lowering the false positive rates by 42.9%, which is very important for user trust. The improved claim extraction module gets 91% of the verifiable claims correct, thus, it is a significant step towards lowering the verification noise. In addition, this study introduces MVS-AI provides significant improvements in terms of factual accuracy, reduces the number of false-positive results, and increases the likelihood that the assertions made will be reliable by providing a novel framework for detecting hallucinations from multiple sources using a combination of the methods of claim decomposition, semantic embeddings, natural language inference and AI-enhanced verification in Large Language Models.

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